

Common information

August 2024

Theorem 1. *We can communicate X_1 and X_2 using shared and nonshared channels with respective rates r_{12}, r_1, r_2 iff there exists a joint (Y_{12}, X_1, X_2) (that is, with the given marginal on (X_1, X_2)) satisfying the following three conditions:*

$$r_{12} \geq I(Y_{12}; (X_1, X_2)), \quad r_1 \geq H(X_1 | Y_{12}), \quad r_2 \geq H(X_2 | Y_{12}).$$

Proof of $\implies$. We show that the above condition is necessary: that if there is a coding scheme with rates r_{12}, r_1, r_2 , then there is a (Y_{12}, X_1, X_2) satisfying the above inequalities.

We can view a coding scheme that achieves these rates as a probability space with inputs $(X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^m, X_2^m)$, shared message Z_{12} , and non-shared messages Z_1, Z_2 , all random variables. The idea is to use these to construct a joint (X_1, X_2, Y_{12}) satisfying the desired inequalities. We construct these random variables as follows:

- Sample $(X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^m, X_2^m)$; compute the corresponding messages Z_{12}, Z_1, Z_2 .
- Sample a uniformly random $k \in [m]$ — this will be used as an index.
- Set $(X_1, X_2) = (X_1^k, X_2^k)$ and $Y_{12} = (Z_{12}, (X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^{k-1}, X_2^{k-1}))$.¹

Note that this indeed gives the desired marginal distribution on (X_1, X_2) . We now check that Y_{12} indeed does not inform too much about (X_1, X_2) :

$$\begin{aligned} I(Y_{12}; (X_1, X_2)) &= H((X_1, X_2)) - H((X_1, X_2) | Y_{12}) \\ &= H((X_1, X_2)) - \sum_{j \in [m]} \mathbb{P}(k = j) H((X_1, X_2) | Y_{12}, k = j) \\ &= H((X_1, X_2)) - \frac{1}{m} \sum_{j \in [m]} H((X_1^j, X_2^j) | Z_{12}, (X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^{j-1}, X_2^{j-1})) \\ &= \frac{1}{m} H(((X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^m, X_2^m))) - \frac{1}{m} H(((X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^m, X_2^m)) | Z_{12}) \\ &= \frac{1}{m} I(((X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^m, X_2^m)); Z_{12}) \\ &\leq \frac{1}{m} H(Z_{12}) \leq R_{12}. \end{aligned}$$

¹In case you're worried about there being a type error in the definition of Y_{12} : we are assuming that all random variables are discrete, so think of us as simply making a distinct element in the domain of Y_{12} for each value of k , Z_{12} , and $(X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^{k-1}, X_2^{k-1})$.

And we check that telling Y_{12} doesn't leave X_1 with too much entropy:

$$\begin{aligned}
H(X_1|Y_{12}) &= \sum_{j \in [m]} \mathbb{P}(k = j) H(X_1|Y_{12}, k = j) \\
&= \frac{1}{m} \sum_{j \in [m]} H(X_1^j|Z_{12}, (X_1^1, X_2^1), (X_1^2, X_2^2), \dots, (X_1^{j-1}, X_2^{j-1})) \\
&\leq \frac{1}{m} \sum_{j \in [m]} H(X_1^j|Z_{12}, X_1^1, X_1^2, \dots, X_1^{j-1}) \\
&= \frac{1}{m} H((X_1^1, X_1^2, \dots, X_1^m)|Z_{12}) \\
&\leq \frac{1}{m} H(Z_1|Z_{12}) \leq \frac{1}{m} H(Z_1) \leq r_1.
\end{aligned}$$

Completely analogously, $H(X_2|Y_{12}) \leq r_2$. This finishes the proof that the joint distribution (Y_{12}, X_1, X_2) indeed satisfies the desired inequalities, and hence the proof that there exists such a joint distribution. $\square$

Proof of $\Leftarrow$. We now show that the condition is also sufficient — that for any joint (Y_{12}, X_1, X_2) , there is a communication scheme with rates $r_{12} = I(Y_{12}; (X_1, X_2))$. This is implied by the rate $\square$